Table 1: L2 (TDOST classes), all classes — comparable to TDOST / DeepCASAS; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.79	0.45	0.88	0.53	0.76	0.33	0.81	0.44
k-means	Sensor+ToD hist.	0.72	0.31	0.80	0.31	0.84	0.40	0.79	0.34
k-means	MLM	0.80	0.45	**0.89**	**0.55**	0.76	0.35	0.82	**0.45**
k-means	MLM-time	**0.81**	**0.47**	0.89	0.50	0.78	0.33	**0.82**	0.43
k-means	MLM + ToD	0.72	0.31	0.80	0.31	**0.86**	**0.43**	0.79	0.35
k-means	MLM-time + ToD	0.72	0.31	0.78	0.28	**0.86**	0.41	0.79	0.33
SCAN	MLM	0.77	0.34	0.85	0.42	0.79	0.35	0.80	0.37
SCAN	MLM-time	0.75	0.30	0.88	**0.55**	0.77	0.32	0.80	0.39
DEC	MLM	0.78	0.35	0.88	0.46	0.77	0.33	0.81	0.38
DEC	MLM-time	0.77	0.34	**0.89**	0.47	0.75	0.29	0.81	0.37
$K = 100$ *clusters*									
k-means	Sensor hist.	0.81	0.49	**0.89**	0.59	0.79	0.36	0.83	0.48
k-means	Sensor+ToD hist.	0.82	0.50	0.88	0.52	0.87	**0.49**	0.86	**0.51**
k-means	MLM	0.82	0.50	**0.89**	0.59	0.81	0.43	0.84	**0.51**
k-means	MLM-time	0.81	0.50	0.89	0.57	0.81	0.45	0.84	**0.51**
k-means	MLM + ToD	**0.82**	0.50	**0.89**	0.50	**0.89**	0.47	**0.87**	0.49
k-means	MLM-time + ToD	**0.82**	**0.51**	0.88	0.54	0.88	0.47	0.86	**0.50**
SCAN	MLM	0.80	0.45	**0.89**	0.60	0.82	0.40	0.84	0.48
SCAN	MLM-time	0.81	0.47	**0.89**	**0.60**	0.82	0.45	0.84	**0.51**
DEC	MLM	0.80	0.45	0.87	0.54	0.79	0.36	0.82	0.45
DEC	MLM-time	0.80	0.43	**0.89**	0.55	0.80	0.38	0.83	0.45

Table 2: L2, rare classes masked from the average; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.80	0.56	0.88	0.78	0.76	0.38	0.81	0.58
k-means	Sensor+ToD hist.	0.72	0.38	0.81	0.50	0.84	0.47	0.79	0.45
k-means	MLM	0.80	0.57	**0.89**	**0.81**	0.77	0.40	0.82	**0.59**
k-means	MLM-time	**0.81**	**0.59**	0.89	0.80	0.78	0.39	**0.83**	**0.59**
k-means	MLM + ToD	0.73	0.38	0.80	0.49	**0.86**	**0.51**	0.80	0.46
k-means	MLM-time + ToD	0.73	0.39	0.78	0.45	**0.86**	0.48	0.79	0.44
SCAN	MLM	0.78	0.42	0.85	0.58	0.79	0.41	0.81	0.47
SCAN	MLM-time	0.76	0.37	0.88	0.78	0.78	0.38	0.81	0.51
DEC	MLM	0.79	0.43	0.89	0.74	0.78	0.38	0.82	0.52
DEC	MLM-time	0.78	0.43	**0.89**	0.75	0.76	0.34	0.81	0.51
$K = 100$ *clusters*									
k-means	Sensor hist.	0.82	0.61	**0.89**	**0.80**	0.79	0.41	0.83	0.61
k-means	Sensor+ToD hist.	0.82	0.61	0.88	0.77	0.88	0.52	0.86	0.63
k-means	MLM	0.82	**0.62**	**0.90**	**0.80**	0.81	0.44	0.84	0.62
k-means	MLM-time	0.81	**0.62**	0.89	0.80	0.81	0.44	0.84	0.62
k-means	MLM + ToD	**0.83**	**0.62**	**0.90**	**0.80**	**0.90**	**0.55**	**0.87**	**0.66**
k-means	MLM-time + ToD	**0.83**	**0.62**	0.88	0.79	0.89	0.53	0.87	0.65
SCAN	MLM	0.81	0.57	**0.90**	0.79	0.82	0.46	0.84	0.61
SCAN	MLM-time	0.81	0.58	**0.89**	0.79	0.82	0.44	0.84	0.61
DEC	MLM	0.80	0.56	0.87	0.74	0.80	0.42	0.83	0.57
DEC	MLM-time	0.80	0.54	**0.89**	0.76	0.81	0.44	0.83	0.58

Table 3: L1 (raw CASAS activities), rare classes masked from the average; $m = all$ labelled windows per cluster. W/M = weighted/macro F1.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W-F1	M-F1	W-F1	M-F1	W-F1	M-F1	W-F1	M-F1
$K = 20$ *clusters*									
k-means	Sensor hist.	0.79	0.58	0.87	0.60	0.29	0.26	0.65	0.48
k-means	Sensor+ToD hist.	0.70	0.38	0.80	0.38	0.72	0.44	0.74	0.40
k-means	MLM	0.80	**0.59**	**0.89**	**0.65**	0.30	0.26	0.66	**0.50**
k-means	MLM-time	**0.80**	**0.59**	**0.88**	**0.65**	0.30	0.27	0.66	**0.50**
k-means	MLM + ToD	0.73	0.37	0.80	0.37	0.72	**0.45**	**0.75**	0.40
k-means	MLM-time + ToD	0.72	0.37	0.78	0.32	**0.74**	**0.45**	**0.74**	0.38
SCAN	MLM	0.77	0.47	0.84	0.46	0.32	0.26	0.64	0.40
SCAN	MLM-time	0.75	0.44	0.87	0.60	0.34	0.28	0.65	0.44
DEC	MLM	0.78	0.51	0.87	0.58	0.29	0.25	0.65	0.45
DEC	MLM-time	0.77	0.47	**0.89**	0.61	0.26	0.22	0.64	0.43
$K = 100$ *clusters*									
k-means	Sensor hist.	0.81	0.62	**0.89**	0.65	0.32	0.30	0.67	0.52
k-means	Sensor+ToD hist.	0.81	0.63	0.87	0.63	0.76	**0.57**	0.82	0.61
k-means	MLM	0.81	**0.63**	**0.89**	0.65	0.33	0.32	0.68	0.53
k-means	MLM-time	0.81	0.62	0.88	0.64	0.34	0.33	0.68	0.53
k-means	MLM + ToD	0.82	0.63	**0.89**	**0.66**	**0.78**	**0.58**	**0.83**	**0.62**
k-means	MLM-time + ToD	**0.82**	**0.63**	0.88	0.64	0.77	0.56	**0.83**	0.61
SCAN	MLM	0.80	0.59	**0.89**	0.64	0.34	0.33	0.68	0.52
SCAN	MLM-time	0.81	0.60	**0.89**	0.64	0.35	0.34	0.68	0.53
DEC	MLM	0.80	0.58	0.87	0.59	0.30	0.28	0.66	0.48
DEC	MLM-time	0.79	0.56	**0.89**	0.63	0.32	0.30	0.67	0.50